

DS&AI

Machine Learning

DPP 02

Unsupervised Learning

Q1 Consider the following Confusion Matrix:

	Predicted A	Predicted B	Predicted C
Actual A	40	5	5
Actual B	10	30	10
Actual C	5	10	35

- (A) Precision = 0.73
Recall = 0.8
- (B) Precision = 0.73
Recall = 0.76
- (C) Precision = 0.76
Recall = 0.81
- (D) Precision = 0.8
Recall = 0.8

Q2 Assume, you want to cluster 7 observations into 3 clusters using the K-Means clustering algorithm. After the first iteration, the clusters:

C1, C2, C3 have the following observations

C1: ((1,1), (4,4), (7,7))

C2: {(0,4), (4,0)}

C3: {(5,5), (9,9)}

What will be the cluster centroids if you want to proceed for the second iteration?

- (A) Ct: (4,4), C2: (2,2), C3: (7,7)
- (B) Ct: (2,2), C2: (0,0), C3: (5,5)
- (C) Ct: (6,6), C2: (4,4), C3: (9,9)
- (D) None of these

Q3 We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not studied.

GPA	Studied	Passed
L	F	F
L	T	T

M	F	F
M	T	T
H	F	T
H	T	T

What is the entropy $H(\text{Passed} \mid \text{GPA})$ and $H(\text{Passed} \mid \text{Studied})$?

- (A) $H(\text{Passed} \mid \text{GPA}) = 0.66$
 $H(\text{Passed} \mid \text{Studied}) = 0.46$
- (B) $H(\text{Passed} \mid \text{GPA}) = 0.46$
 $H(\text{Passed} \mid \text{Studied}) = 0.66$
- (C) $H(\text{Passed} \mid \text{GPA}) = 0.46$
 $H(\text{Passed} \mid \text{Studied}) = 0.92$
- (D) $H(\text{Passed} \mid \text{GPA}) = 0.33$
 $H(\text{Passed} \mid \text{Studied}) = 0.66$

Q4 Given Six points with the following attributes:

Point	x coordinate	y coordinate
p1	0.4005	0.5306
p2	0.2148	0.3854
p3	0.3457	0.3156
p4	0.2652	0.1875
p5	0.0789	0.4139
p6	0.4548	0.3022

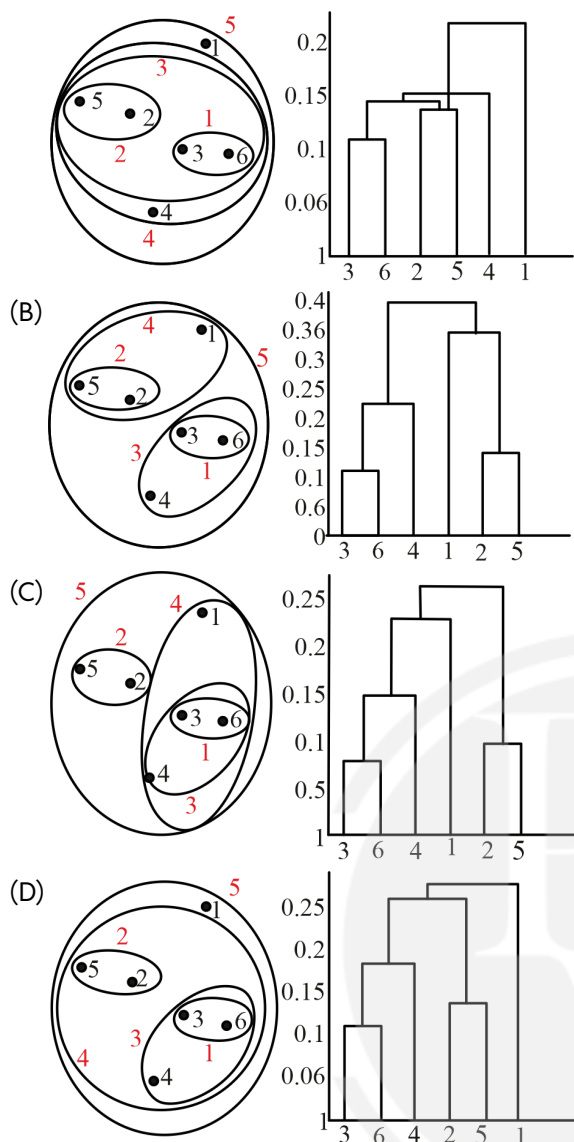
Table : X-Y coordinates of six points

	p1	p2	p3	p4	p5	p6
p1	0.0000	0.2357	0.2218	0.3688	0.3421	0.2347
p2	0.2357	0.0000	0.1483	0.2042	0.1388	0.2540
p3	0.2218	0.1483	0.0000	0.1513	0.2843	0.1100
p4	0.3688	0.2042	0.1513	0.0000	0.2932	0.2216
p5	0.3421	0.1388	0.2843	0.2932	0.0000	0.3921
p6	0.2347	0.2540	0.1100	0.2216	0.3921	0.0000

Q5 Which of the following clustering representations and dendrogram depicts the use of MIN or Single link proximity function in hierarchical clustering?

(A)



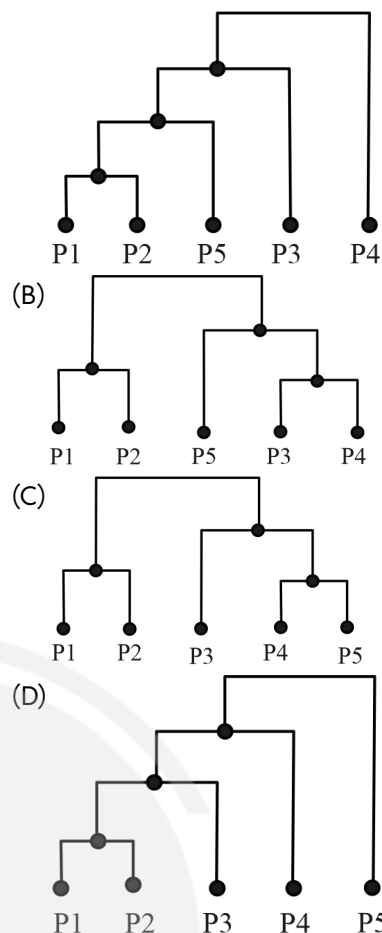


Q6 Consider the distance matrix in Table1 to perform the complete link hierarchical clustering.

	P1	P2	P3	P4	P5
P1	0.00	0.10	0.41	0.55	0.35
P2	0.10	0.00	0.64	0.47	0.98
P3	0.41	0.64	0.00	0.44	0.85
P4	0.55	0.47	0.44	0.00	0.76
P5	0.35	0.98	0.85	0.76	0.00

Which ONE of the following is the correct representation of the clusters produced using dendrogram?

(A)



Q7 Given the following 8 data points in a 2-dimensional space:

Point	Coordinates (x, y)
A1	(2, 10)
A2	(2, 5)
A3	(8, 4)
A4	(5, 8)
A5	(7, 5)
A6	(6, 4)
A7	(1, 2)
A8	(4, 9)

Assume that initial centroid:

Centroid 1 (C1): A1 = (2, 10)

Centroid 2 (C2): A4 = (5, 8)

Centroid 3 (C3): A7 = (1, 2)

Cluster these points into 3 clusters using the K-means algorithm.

Choose the correct option.

(A) Cluster 1: {A1, A2}

Cluster 2: {A3, A4, A5, A6, A8}

Cluster 3: {A7}



- (B) Cluster 1: {A1}
Cluster 2: {A3, A4, A5, A6, A8}
Cluster 3: {A2, A7}
- (C) Cluster 1: {A1, A2}
Cluster 2: {A3, A4, A5, A8}
Cluster 3: {A6, A7}
- (D) Cluster 1: {A1}
Cluster 2: {A4, A5, A6, A8}
Cluster 3: {A2, A3, A7}

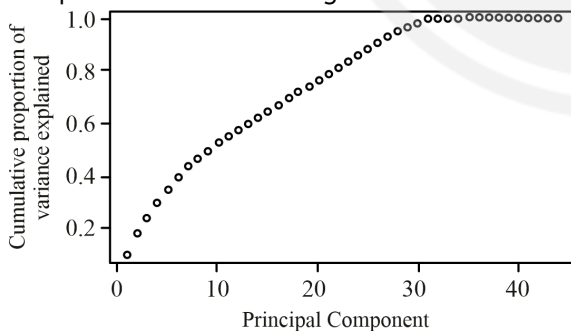
Q8 We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

What is the entropy $H(\text{Passed})$?

- (A) 0.46 (B) 0.63
(C) 0.92 (D) 0.23

Q9 What are the optimal number of principal components in the below figure?



- (A) 10 (B) 20
(C) 30 (D) 40

Q10 Problem Given the dataset {a, b, c, d, e} and the following distance matrix, construct a dendrogram by complete linkage hierarchical clustering using the agglomerative method.

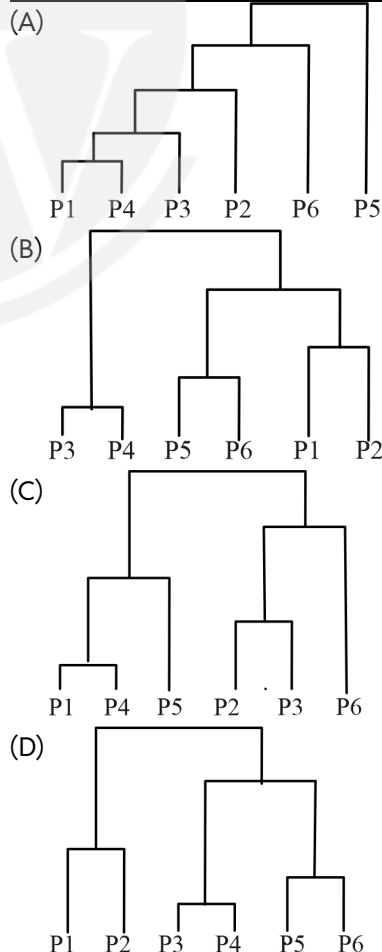
	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10

c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

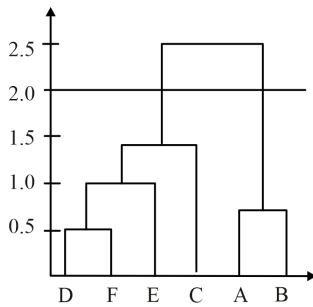
- Q11** Consider the following one dimensional data set: 12, 22, 2, 3, 33, 27, 5, 16, 6, 31, 20, 37, 8, 18. Given $k = 3$ and initial cluster centers to be 5, 6 and 31, what are the final cluster centers obtained on applying the k-means algorithm?
- (A) 5, 18, 30 (B) 5, 18, 32
(C) 6, 19, 32 (D) 4.8, 17.6, 32

Q12 For the similarity matrix given below, show the hierarchy of clusters created by the single link clustering algorithm.

	P1	P2	P3	P4	P5	P6
P1	1.0000	0.7895	0.1579	0.0100	0.5292	0.3542
P2	0.7895	1.0000	0.3684	0.2105	0.7023	0.5480
P3	0.1579	0.3684	1.0000	0.8421	0.5292	0.6870
P4	0.0100	0.2105	0.8421	1.0000	0.3840	0.5573
P5	0.5292	0.7023	0.5292	0.3840	1.0000	0.8105
P6	0.3542	0.5480	0.6870	0.5573	0.8105	1.0000



- Q13** In the figure below, if you draw a horizontal line on y-axis for $y=2$. What will be the number of clusters formed?



- (A) 1 (B) 2
(C) 3 (D) 4
- Q14** If two variables V1 and V2 are used for clustering. Which of the following are true for K-means clustering with $k=3$?

1. If V1 and V2 has a correlation of 1, the cluster centroids will be in a straight line
2. If V1 and V2 has a correlation of 0, the cluster centroids will be in straight line

Choose the correct answer?

- (A) 1 Only
(B) 2 Only
(C) Both 1 and 2
(D) None of the above
- Q15** Which of the following clustering algorithms is the most sensitive to outliers?
- (A) K-means clustering algorithm
(B) K-medians clustering algorithm
(C) K-modes clustering algorithm
(D) None of the above
- Q16** Suppose you run k-means with $k=2$ and have the initial cluster centers be $\mu_1 = (4, 0)$ and $\mu_2 = (0, 3)$. What will be the final cluster centers when k-means converges? Show your work and explain how you know that k-means has converged.

- Q17** Given the dataset:
(1,2), (2,3), (3,1), (8,8), (9,9), (10,10), (6, 7)
Assuming the initial centroids for $K=3$ means clustering to be: $C_1 = (2, 3)$ $C_2 = (3, 1)$ $C_3 = (9, 9)$
One iteration of the Expectation-Maximization Algorithm for K-means clustering will update C_1 to:

- Q18** Consider the pairwise distances between 5 points given as matrix.

Since the matrix is symmetric, we have given only the lower triangle.

Find the final Clustering using single linkage.

	1	2	3	4	5
1	0				
2	2	0			
3	6	3	0		
4	10	9	7	0	
5	9	8	5	4	0

- Q19** Consider n feature vectors $x \in \mathbb{R}^d$ that need to be projected into a lower-dimensional space $\mathbb{R}^k$ using k principal component basis vectors. The projection matrix W of size $d \times k$ consists of these orthonormal principal component vectors.

Which of the following expressions correctly represents the reconstruction error?

- (A) $\|x - WW^T x\|_2^2$ (B) $\|x - W^T x\|_2^2$
(C) $\|x - W^T W x\|_2^2$ (D) $\|x - W x\|_2^2$

- Q20** In the context of binary classification, if we continuously decrease the classifier threshold, how do the True Positive Rate (TPR) and False Positive Rate (FPR) behave?

- (A) TPR increases while FPR decreases
(B) TPR decreases while FPR increases
(C) Both TPR and FPR increase
(D) Both TPR and FPR decrease

Q21 [Common Data]

Let's do principal components analysis (PCA)!

Consider this sample of six points $x_i \in \mathbb{R}^2$.

$$\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}.$$

Compute the mean of the sample points and write the centered design matrix X .

- Q22** Find all the principal components of this sample. Write them as unit vectors.

- Q23**
- Which of those two principal component would be preferred if you use only one?
 - What information does the PCA algorithm use to decide that one principal component is

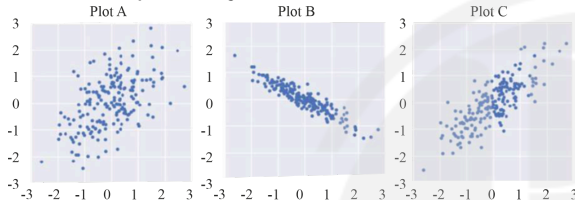


better than another?

- From an optimization point of view, why do we prefer that one?

Q24 Compute the vector projection of each of the original sample point (not the centered sample points) onto your preferred principal component. By “vector projection” we mean that the projected points are still in $\mathbb{R}^2$. (Don't just give us the principal coordinate give us the projected point).

Q25 Suppose we are given three datasets A, B and $C \in \mathbb{R}^{100 \times 2}$ i.e. each dataset consists of 100 data points in two dimensions. We visualize the datasets using scatterplots, labelled A, B, and C, respectively:



If we applied PCA to each of the above dataset and used the first two principal component, which dataset(s) would have to lowest reconstruction error?

- (A) Dataset A
(B) Dataset B
(C) Dataset C
(D) Cannot be determined

Q26 (Hierarchical Clustering) Consider six points in 2D, some of them overlap. We start with three clusters (rather than from scratch):

Cluster 1 = (0,0), (1,1), (2,2)

Cluster 2 = (0,1), (1,0)

Cluster 3 = (0,0)

With Euclidean distance and complete linkage, which clusters are merged in the next step of hierarchical clustering?

Q27 Perform hierarchical clustering with single linkage in one-dimensional space on the following points: -5, -1, 0, 3, 9, 11. Draw the clustering tree.

Iteration 1: group A=(-1, 0): distance 1

Iteration 2: group B=(9, 11): distance 2

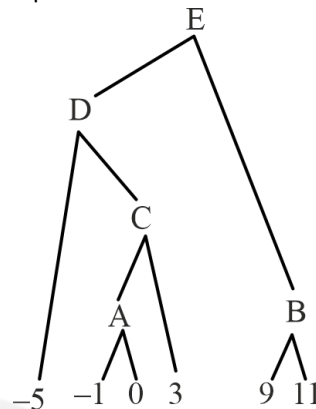
Iteration 3: group C=(A, 3): distance 3 between 3 and the element 0 in A
Iteration 4: group D=(-5,

C): distance 4 between -5 and the element -1 in C

Iteration 5: group E=(D, B): distance 6 between elements 3 and 9.

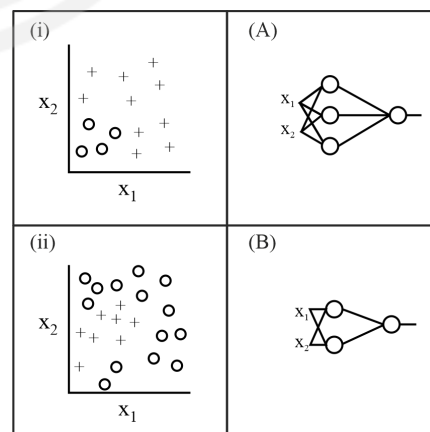
No tree graph, but correct steps: 7 points

Almost correct graph and solution, minor errors: 7 points



Q28 There are five points in one-dimensional space: $a = 1.5$, $b = 2$, $c = 6$, $d = 7$, $e = 9$. Perform Hierarchical Agglomerative Clustering with Single Linkage, with the following extra constraint: at no time should c and d be in the same cluster. Complete the resulting clustering tree diagram (i.e., the dendrogram).

Q29 Recall that the output of a perceptron is 0 or 1. For each of the two following data sets, select the perceptron network with the fewest nodes that will separate the classes. You can use the same network more than once



(A) (i)- A; (ii)-B

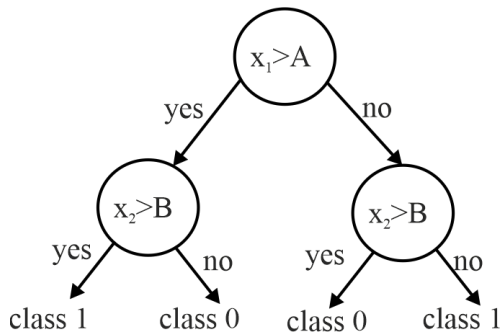
(B) (i)- B; (ii)-A

(C) (i)- A; (ii)-A

(D) (i)- B; (ii)-B

Q30





Assume we have the decision tree in Figure which classifies two dimensional vectors $(X_1, X_2) \in \mathbb{R}^2$. In other words, the values A and B are never used in the inputs. Can this decision tree be implemented as a 1-NN? If so, explicitly say what are the values you use for the 1-NN (you should use the minimal number possible). If not, either explain why or provide a counter example.

Q31 We are dealing with samples x where x is single value. We would like to test two alternative regression models:

1. $y = ax + e$
2. $y = ax + bx^2 + e$

We make the same assumptions we had in class about the distribution of e ($e \sim N(0, \sigma^2)$).

(a) Assume we have n samples: $x_1, \dots, x_n$ with their corresponding y values, $y_1, \dots, y_n$. Derive the value assigned to be in model 2. You can use a in the equation for b .

$$b = \frac{\sum_i y_i x_i^2 - a \sum_i x_i^3}{\sum_i x_i^4}$$

Q32 Which of the two models is more likely to fit the training data better?

- (A) Model 1
 (B) Model 2
 (C) Both will fit equally well
 (D) impossible to tell

Q33 [Common Data] next 4 Ques.

About 2/3 of your email is spam so you downloaded an open source spam filter based on word occurrences that uses the Naive Bayes classifier.

Assume you collected the following regular and spam mails to train the classifier, and only three words are informative for this classification, i.e.,

each email is represented as a 3-dimensional binary vector whose components indicate whether the respective word is contained in the email.

'study'	'free'	'money'	Category
1	0	0	Regular
0	0	1	Regular
1	0	0	Regular
1	1	0	Regular
0	1	0	Spam
0	1	0	Spam
0	1	0	Spam
0	1	0	Spam
0	1	1	Spam
0	1	1	Spam
0	1	1	Spam
0	1	1	Spam

Q34 You find that the spam filter uses a prior $p(\text{spam}) = 0.1$. Explain (in one sentence) why this might be sensible.

Q35 Give the following model parameters when estimated as maximum-likelihood with add-one smoothing (i.e., using pseudo counts of one).

$$P(\text{study} | \text{spam}) = 1/10$$

$$P(\text{study} | \text{regular}) = 2/3$$

$$P(\text{free} | \text{spam}) = 9/10$$

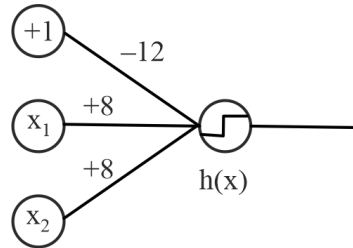
$$P(\text{free} | \text{regular}) = 1/3$$



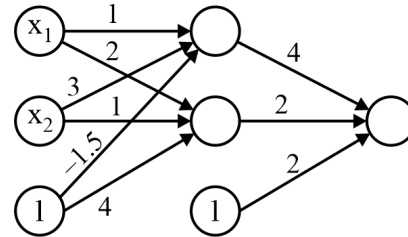
$$P(\text{money} \mid \text{spam}) = 1/2$$

$$P(\text{money} \mid \text{regular}) = 1/3$$

- Q36** Based on the prior and conditional probabilities above, give the model probability $P(\text{spam}/s)$ that the sentence s "money for psychology study" is spam.
- Q37** What should be the value of the prior $p(\text{spam})$ if we would like the above sentence to have the same probability as being spam as not spam, i.e., it would be classified as spam with probability 0.5?
- Q38** Can a neural network be used to model the following machine learning algorithms? If so, state the neural network structure (how many hidden layers are required) and the activation function(s) used at the internal and output nodes. If not, describe why not in one or two sentences.
- Linear regression
 - Logistic regression
 - k-nearest neighbors
- Q39** Which of the following statements are true for k-NN classifiers (circle all answers that are correct).
1. The classification accuracy is better with larger values of k .
 2. The decision boundary is smoother with smaller values of k .
 3. k-NN is a type of instance-based learning.
 4. k-NN does not require an explicit training step.
 5. The decision boundary is linear.
- Q40** Is it possible for a 2-class 1-NN classifier to always classify all new examples as positive even though there are negative examples in the training data? If yes, show an example. If no, briefly explain.
- Q41** What is the output of the network for $x_1=1, x_2=1$?



- Q42** What is the output of the network for $x_1=1, x_2=1$?



- Q43** Below is neural network with weight a, b, c, d, e, f . The input are x_1 and x_2 .

The first hidden layer computes $r_1 = \max(c \cdot x_1 + e \cdot x_2, 0)$ and $r_2 = \max(d \cdot x_1, 0)$ and $r_2 = \max(d \cdot x_1 + f \cdot x_2, 0)$.

The second hidden layer computes $s_1 =$

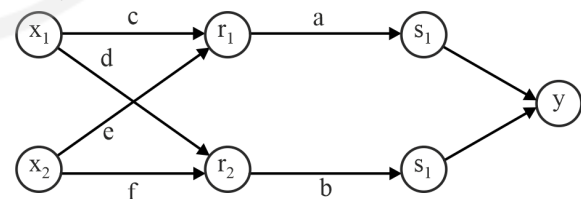
$$\frac{1}{1 + \exp(-a \cdot r_1)} \text{ and } s_2 = \frac{1}{1 + \exp(-a \cdot r_1)}.$$

The output layer computes $y = s_1 + s_2$.

Note that the weight a, b, c, d, e, f and indicate along the edges of the neural network here.

Suppose the network has input $x_1 = 1, x_2 = -1$.

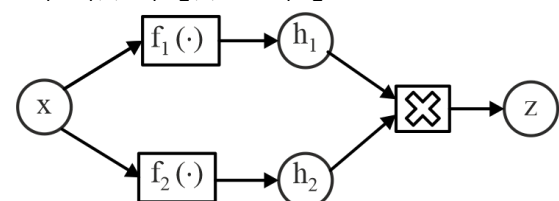
The weight values are $a=1, b=1, c=4, d=1, e=2, f=2$.



- Q44** [Common Data] next 2 Ques.

Backpropagation below is a deep network with input x . Values x, h_1, h_2, z , are all scalars.

$$h_1 = f_1(x), h_1, f_2(x), z = h_1 h_2$$



What of the following options correctly represent the expression

$$\text{of } \frac{\partial z}{\partial h_1}.$$



- (A) h_1 (B) h_2
 (C) $H_1 h_2$ (D) None

Q45 What of the following options correctly represent the expression of $\frac{\partial z}{\partial h_2}$.

- (A) h_1 (B) h_2
 (C) $H_1 h_2$ (D) None

Q46 A neural network has many layers, with the final two being X (with 4 nodes) and Y (with one node).

Four weights $\{w_1 = 0.1, w_2 = -0.5, w_3 = 0.8, w_4 = 0.6\}$ connect x_1, x_2, x_3, x_4 to y_1 .

On the most recent forward pass through the network the outputted values of X are:

$x_1 = 0.3, x_2 = -0.7, x_3 = 0.5, x_4 = 0.9$.

The activation function for node y_1 is:

$y_1 = \frac{1}{1+e^{-s}}$ where s is the sum of weighted inputs into y_1 .

The error function is: $\frac{1}{2}(T - y_1)^2$,

where T is the target value. In this case $T = 0.8$

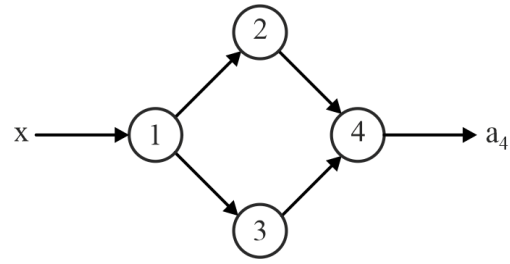
The learning rate is 0.05, and the minibatch size is 1.

Taks word:

- Compute the value of y_1 during the forward pass.
- Compute the gradients of w_1, w_2, w_3, w_4 using backpropagation
- Find the new values for the four weights after the backwards pass.

Q47 Consider the neural network below. Note that:

- x , a scalar, is the input. a_4 , also a scalar, is the output.
- Each a_i value is the final output for neuron i in the network.
- Each z_i value is the pre-activation value for neuron i in the network (i.e. the dot product).
- w_1 is the weight used by neuron 1.
- Let w_{ij} be the weight from neuron i to neuron j .
- The activation function for each neuron is the function $g(x) = \exp(x)$.
- $L(y, a_4)$ is the loss function. It is $(y - a_4)^2$, where y is the training label.



Q48 Which of the options are correct

- (A) Maximum value of local gradient when sigmoid is used as an activation function is 0.5.
 (B) Minimum value of local gradient when sigmoid is used as an activation function is 0.25
 (C) Maximum value of local gradient when tanh is used as an activation function is 1
 (D) Minimum value of local gradient when Relu is used as an activation function is 0

Q49 Which of the following activation functions can lead to vanishing gradients?

- (A) ReLU
 (B) Tanh
 (C) Sigmoid
 (D) All of the above

Q50 Which of the options are correct

- (A) You are solving a binary classification task, and the final two layers in your network are a ReLU activation followed by a sigmoid activation. This leads to all predictions to be positive.
 (B) A 2-layer fully connected neural network with 5 neurons in each layer has a total of 60 parameters (i.e. weights and biases)
 (C) Vanishing gradient causes deeper layers to learn more slowly than earlier layers
 (D) The backpropagated gradient through a tanh non-linearity is always greater than in magnitude than the upstream gradient.

Q51 Which of the following properties, if present in an activation function CANNOT be used in a neural network?

- (A) The function is periodic
 (B) The function is monotonic
 (C) The function is unbounded



(D) Both a and b

Q52 Which of the following is true about the vanishing gradient problem? (Circle all that apply)

(A) Tanh is usually preferred over sigmoid because it doesn't suffer from vanishing gradients

(B) Vanishing gradient causes deeper layers to learn more slowly than earlier layers

(C) Leaky ReLU is less likely to suffer from vanishing gradients than sigmoid

(D) Xavier initialization can help prevent the vanishing gradient problem

(E) None of the above



Answer Key

Q1 (A)

Q2 (A)

Q3 (A)

Q4 0~0

Q5 (A)

Q6 (A)

Q7 (A)

Q8 (C)

Q9 (C)

Q10 0~0

Q11 (A)

Q12 (A)

Q13 (B)

Q14 (A)

Q15 (A)

Q16 $C_1 \Rightarrow \left(\frac{3}{2}, \frac{5}{2}\right)$ Q17 $C_1 \Rightarrow \left(\frac{3}{2}, \frac{5}{2}\right)$

Q18 4~4

Q19 (A)

Q20 (C)

Q21
$$X^T = \begin{bmatrix} -1 & -1 \\ -1 & 0 \\ 0 & -1 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Q22 $\lambda = 6, 2$ Q23 $\lambda = 6, 2$ Q24 ${}_6 \binom{4/2}{4/2}$

Q25 (A)

Q26 $C_2 C_3$

Q27 0~0

Q28 0~0

Q29 (A)

Q30 0~0

Q31 0~0

Q32 (A)

Q33 0~0

Q34 0~0

Q35 0~0

Q36 0~0

Q37 0~0

Q38 0~0

Q39 0~0

Q40 0~0

Q41 0~0

Q42 0~0

Q43 0~0

Q44 (A)

Q45 (A)

Q46 0~0

Q47 0~0

Q48 (A)

Q49 (A)

Q50 (A)

Q51 (A)

Q52 (A)


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